```sql
/* ============================================================
   SWIGGY FOOD DELIVERY DATA ANALYSIS
   ------------------------------------------------------------
   SQL Data Analysis Project | MySQL
   ------------------------------------------------------------
   Data Import • Data Cleaning • Data Validation
   Dimensional Data Modeling • KPI Analysis
   Business Analysis • Trend Analysis • Rating Analysis
   ============================================================ */
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```


```sql
Drop table  swiggy;
CREATE Table Swiggy(
                    State VARCHAR(60),
                    City VARCHAR(50),
                    Order_Date     VARCHAR(50),
```

```sql
                    Restaurant_Name VARCHAR(70),
                    Location VARCHAR(70),
                    Category VARCHAR(70),
                    Dish_Name     VARCHAR(300),
                    Price_INR       int,
                    Rating decimal(10,2),
                    Rating_Count int
);


LOAD DATA INFILE 'D:/New folder (3)/Uploads/Swiggy_Data.csv'
INTO TABLE swiggy
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS;

-- we inset date as a Varchar because IN csv file
-- we don't modify anything that's why i first add
-- using varchar then i change it data Type
UPDATE swiggy
SET Order_Date = STR_TO_DATE(Order_Date, '%d-%m-%Y');

describe swiggy;

ALTER TABLE swiggy
```

```
MODIFY Order_Date DATE;


-- Data Validation and Cleaning

-- Null Check

select

	SUM(CASE When State IS NULL THEN 1 ELSE 0 END) AS  STATE_NULl,

   SUM(CASE When City IS NULL THEN 1 ELSE 0 END) AS  City_NULl,

   SUM(CASE When Order_Date IS NULL THEN 1 ELSE 0 END) AS  Order_date_NULl,

   SUM(CASE When Restaurant_Name IS NULL THEN 1 ELSE 0 END) AS
Restaruant_Name_NULl,

   SUM(CASE When Location IS NULL THEN 1 ELSE 0 END) AS  Location_NULl,

   SUM(CASE When Category IS NULL THEN 1 ELSE 0 END) AS  Category_NULl,

   SUM(CASE When Dish_Name IS NULL THEN 1 ELSE 0 END) AS  Dish_Name_NULl,

   SUM(CASE When Price_INR IS NULL THEN 1 ELSE 0 END) AS  Price_NULl,

	SUM(CASE When Rating IS NULL THEN 1 ELSE 0 END) AS  Rating_NULl,

	SUM(CASE When Rating_Count IS NULL THEN 1 ELSE 0 END) AS  Ratting_COunt_NULl

FROM swiggy;


-- Blank or Empty check

Select * From swiggy

Where

State='' OR City='' OR Restaurant_Name='';


-- Duplicates Detections

SELECT State, City , Order_Date,Restaurant_Name,

Location , Category, Dish_Name,Price_INR,Rating,Rating_Count,
```

```sql
COUNT(*) as CNT
from swiggy
GROUP BY
State, City , Order_Date,Restaurant_Name,
Location , Category, Dish_Name,Price_INR,Rating,Rating_Count
Having Count(*)>1 order by Count(*)>1 DESC;


-- Delete Duplicates

WITH CTE AS (
   SELECT
      id,
      ROW_NUMBER() OVER(
         PARTITION BY State, City, Order_Date, Restaurant_Name,
                  Location, Category, Dish_Name,
                  Price_INR, Rating, Rating_Count
         ORDER BY id
      ) AS rn
   FROM swiggy
)
DELETE s
FROM swiggy s
JOIN CTE c
ON s.id = c.id
WHERE c.rn > 1;
```

```
-- Creating Schema
-- dimensioal Table
-- Date Table
CREATE TABLE dim_date(
        date_id INT AUTO_INCREMENT PRIMARY KEY,
  Full_Date Date,
  Year int,
  Month INT ,
  Month_Name VARCHAR(30),
  Quarter INT,
  Day_Num INT,
  Week_Num INT
  )
select * from dim_Date;



-- dim location
CREATE TABLE dim_Location (
            location_id INT AUTO_INCREMENT PRIMARY KEY,
            State VARCHAR(200),
    City VARCHAR(200),
    Location VARCHAR(300)
    );
```

```sql
-- Restaurant Table
CREATE TABLE dim_restaurant(
                    Restaurant_id INT AUTO_INCREMENT PRIMARY KEY,
                    Restaruant_Name VARCHAR(300)
      );


-- dim_Dish
CREATE TABLE dim_dash(
          dish_id INT AUTO_INCREMENT PRIMARY KEY,
   Dish_Name VARCHAR(300)
   );

-- dim_Category
CREATE TABLE dim_Category (
          Category_id INT AUTO_INCREMENT PRIMARY KEY,
   Category VARCHAR(300)
   );

-- Fact Table
CREATE TABLE Fact_Swiggy_orders(
          order_id INT AUTO_INCREMENT PRIMARY KEY,
   date_id int,
   Price_INT DECIMAL(10,2),
   Rating DECIMAL(4,2),
   Rating_Count INT,
```

```
   location_ID INT,
   restaurant_ID INT,
   category_ID INT ,
   Dish_ID INT,

        FOREIGN KEY( Date_id) REFERENCES dim_date(date_id),
   FOREIGN KEY( location_ID) REFERENCES dim_location(location_id),
   FOREIGN KEY( restaurant_ID) REFERENCES dim_restaurant(restaurant_id),
   FOREIGN KEY( category_ID) REFERENCES dim_category(category_id),
   FOREIGN KEY( Dish_ID) REFERENCES dim_dash(dish_id)
   );

-- Inset data
INSERT INTO dim_date
(
   Full_Date,
   Year,
   Month,
   Month_Name,
   Quarter,
   Day_Num,
   Week_Num
)
SELECT DISTINCT
   Order_Date,
   YEAR(Order_Date),
```

```
    MONTH(Order_Date),
    MONTHNAME(Order_Date),
    QUARTER(Order_Date),
    DAY(Order_Date),
    WEEK(Order_Date)
FROM swiggy
WHERE Order_Date IS NOT NULL;

-- dim_location
INSERT INTO dim_location (State, City, Location)
SELECT DISTINCT
    State,
    City,
    Location
FROM swiggy;

-- dim_restaurant
INSERT INTO dim_restaurant (Restaruant_Name)
SELECT DISTINCT
    Restaurant_Name
FROM swiggy;


-- dim_category
INSERT INTO dim_category(Category)
SELECT DISTINCT
```

```
    Category
FROM swiggy;

-- dim_dish
INSERT INTO dim_dash (Dish_Name)
SELECT DISTINCT
    Dish_Name
FROM swiggy;



INSERT INTO Fact_Swiggy_orders(
    date_id,
    Price_INT,
    Rating ,
    Rating_Count,
    location_ID,
    restaurant_ID,
    category_ID,
    Dish_ID
)
SELECT
    dd.date_id,
    s.Price_INR,
    s.Rating,
    s.Rating_Count,
```

```sql
    dl.location_id,
    dr.restaurant_id,
    dc.category_id,
    dsh.dish_id
FROM swiggy s

JOIN dim_date dd
    ON dd.Full_Date = s.Order_Date

JOIN dim_location dl
    ON dl.State = s.State
    AND dl.City = s.City
    AND dl.Location = s.Location
JOIN dim_restaurant dr
    ON dr.Restaruant_Name = s.Restaurant_Name
JOIN dim_category dc
    ON dc.Category = s.Category

JOIN dim_dash dsh
    ON dsh.Dish_Name = s.Dish_Name;



SELECT * FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
JOIN dim_location l ON f.location_id = l.location_id
JOIN dim_restaurant r ON f.restaurant_id = r.restaurant_id
```

```sql
JOIN dim_category c ON f.category_id = c.category_id
JOIN dim_dash di ON f.dish_id = di.dish_id;

-- kPI's
-- Total Orders
SELECT COUNT(*) AS Total_Orders FROM fact_swiggy_orders;

-- Total_Revenue
Select sum(PRICE_INT) AS Total_Revenue FROM fact_swiggy_orders;

-- Average Dist Price
Select avg(PRICE_INT) AS avg_dish_price FROM fact_swiggy_orders;

-- avg_rating
Select avg(rating) AS avg_rating FROM fact_swiggy_orders;



-- Deep Dive Bussniess Analysis
select * from  fact_swiggy_orders;
select * from  dim_date;
select
d.Year,
d.month,
d.month_name,
count(*) as Total_Orders
from fact_swiggy_orders f
```

```
JOIN dim_date d on f.date_id = d.date_id
GROUP BY
d.Year,
d.month_name,
d.month
Order By count(*) DESC;

-- Quanter
select
d.Year,
d.quarter,
count(*) as Total_Orders
from fact_swiggy_orders f
JOIN dim_date d on f.date_id = d.date_id
GROUP BY
d.Year,
d.quarter;

-- Yearly Trend
select
d.Year,
count(*) as Total_Orders
from fact_swiggy_orders f
JOIN dim_date d on f.date_id = d.date_id
GROUP BY
d.Year;
```

```sql
-- Orders by Day of Week (Mon-Sun)
SELECT
    DAYNAME(d.full_date) AS day_name,
    COUNT(*) AS total_orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
GROUP BY DAYNAME(d.full_date), WEEKDAY(d.full_date)
ORDER BY WEEKDAY(d.full_date);

-- Top 10 city by order volumne

SELECT
l.city,
count(*) as Total_Orders
FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_ID
GROUP BY l.city
ORDER BY count(*) DESC Limit 10;

-- Top 10 city by Total_Sales
SELECT
l.city,
sum(f.Price_INT) as Total_Revnue
```

```
FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_ID
GROUP BY l.city
ORDER BY sum(f.Price_INT) DESC Limit 10;


-- Top 10 STATE by Total_Sale
SELECT
l.State,
sum(f.Price_INT) as Total_Revnue
FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_ID
GROUP BY l.State
ORDER BY sum(f.Price_INT) DESC Limit 10;

-- Top 10 Resturant  by Total_Sale
SELECT
r.Restaruant_Name,
sum(f.Price_INT) as Total_Revnue
FROM fact_swiggy_orders f
JOIN dim_restaurant r
ON r.Restaurant_id = f.Restaurant_id
GROUP BY r.Restaruant_Name
ORDER BY sum(f.Price_INT) DESC Limit 10;
```

```sql
-- Top Categories by Order Volume
SELECT
    c.category,
    COUNT(*) AS total_orders
FROM fact_swiggy_orders f
JOIN dim_category c
ON f.category_id = c.category_id
GROUP BY c.category
ORDER BY total_orders DESC limit 10;


-- MySQL version
SELECT
    d.dish_name,
    COUNT(*) AS order_count
FROM fact_swiggy_orders f
JOIN dim_dash d ON f.Dish_ID = d.dish_id
GROUP BY d.dish_name
ORDER BY order_count DESC
LIMIT 10;


-- Cuisine Performance (Orders + Avg Rating) - MySQL
SELECT
```

```
    c.category,
    COUNT(*) AS total_orders,
    AVG(f.rating) AS avg_rating
FROM fact_swiggy_orders f
JOIN dim_category c ON f.category_id = c.category_id
GROUP BY c.category
ORDER BY total_orders DESC;


-- Rating Count Distribution (1-5)
SELECT
    rating,
    COUNT(*) AS rating_count
FROM fact_swiggy_orders
GROUP BY rating
ORDER BY Count(*) DESC;
```